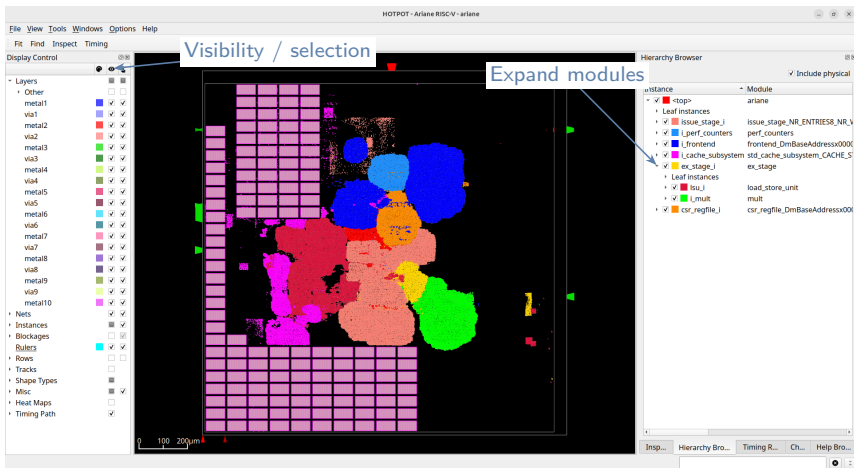
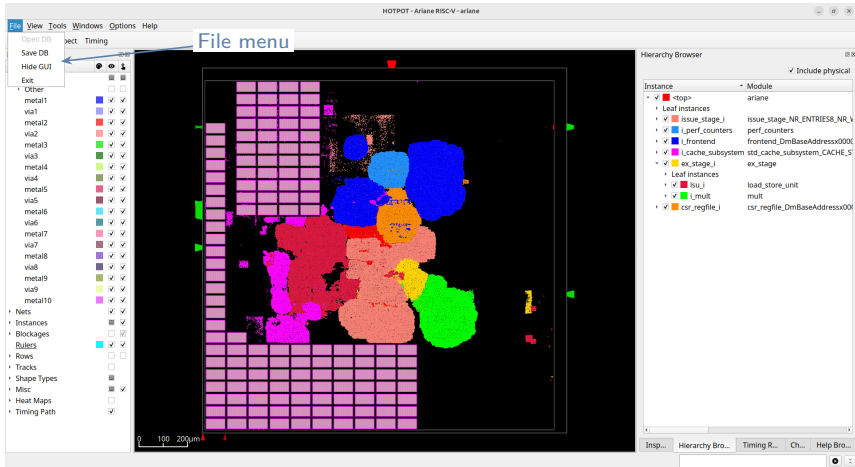


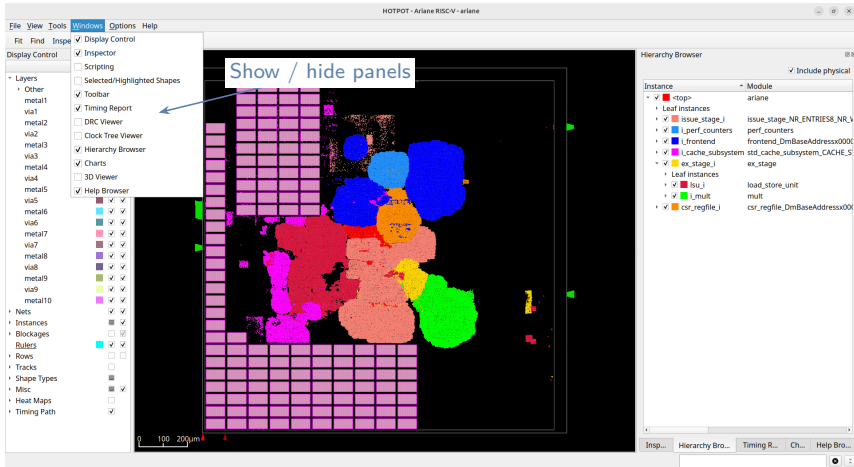
Explore Ariane: a public RISC-V core



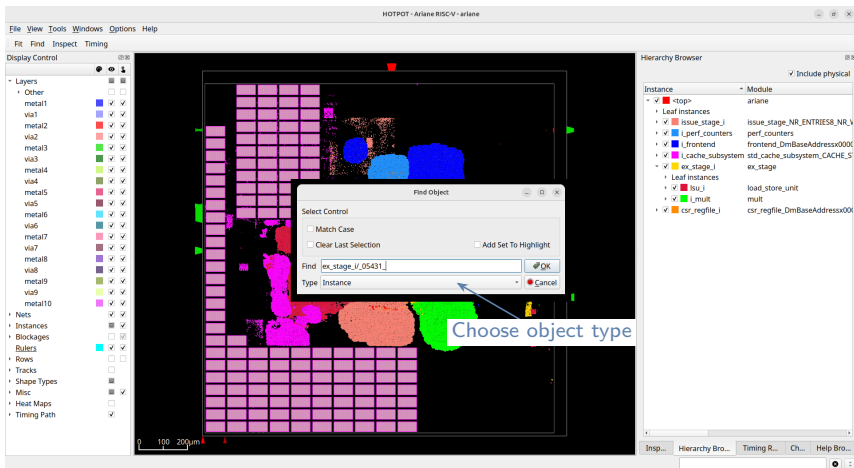
File: open and save databases



Windows: choose the panels



Find an instance by name



Instance: master, placement, terminals

The screenshot displays the HOTPOT - Ariane RISC-V - ariane software interface. The main window shows a circuit layout with various components and layers. The Inspector panel on the right provides detailed information about the selected instance.

Library cell

Instance terminals

Inspector

Name	Value
Type	Inst
Name	ex_stage_IV_05431_
Block	ariane
Module	ex_stage_IV
Master	BBP_X1
Description	Buffer
Placement status	PLACED
Source type	NONE
Dont Touch	False
Orientation	R0
X	805.6 μm
Y	639.8 μm
Terminals	4 items
A	alu_valid_id_ex
Z	ex_stage_IV_00012_
VDD	<none>
VSS	<none>
Timing/Power	ex_stage_IV_05431_
Obstructions	5 items
metal1	Box of MASTER: (805.645, 640.63), (805.645, 640.63), (805.645, 640.63), (805.645, 640.63)
metal1	Box of MASTER: (805.645, 640.63), (805.645, 640.63), (805.645, 640.63), (805.645, 640.63)
metal1	Box of MASTER: (805.645, 640.63), (805.645, 640.63), (805.645, 640.63), (805.645, 640.63)
metal1	Box of MASTER: (805.645, 640.63), (805.645, 640.63), (805.645, 640.63), (805.645, 640.63)
metal1	Box of MASTER: (805.645, 640.63), (805.645, 640.63), (805.645, 640.63), (805.645, 640.63)
BBBox	(805.6, 639.8), (806.17, 641.2)
BBBox Width, Height	(0.57, 1.4)

ITerm: follow the connections



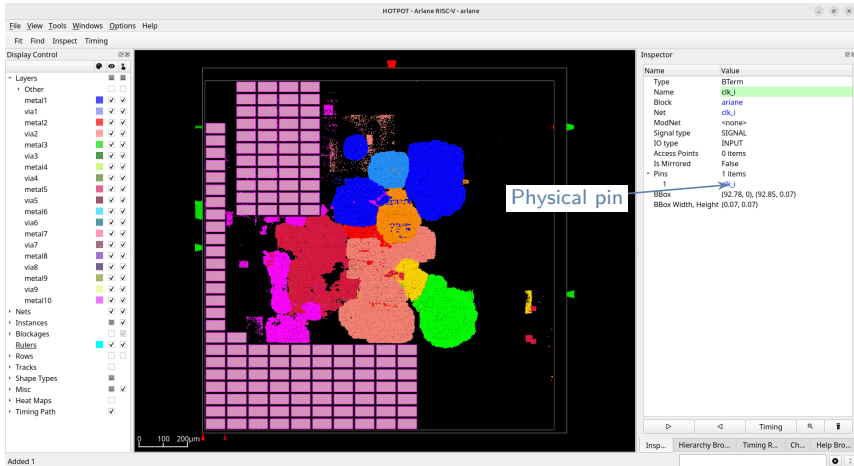
Net: inspect its endpoints

Click an endpoint

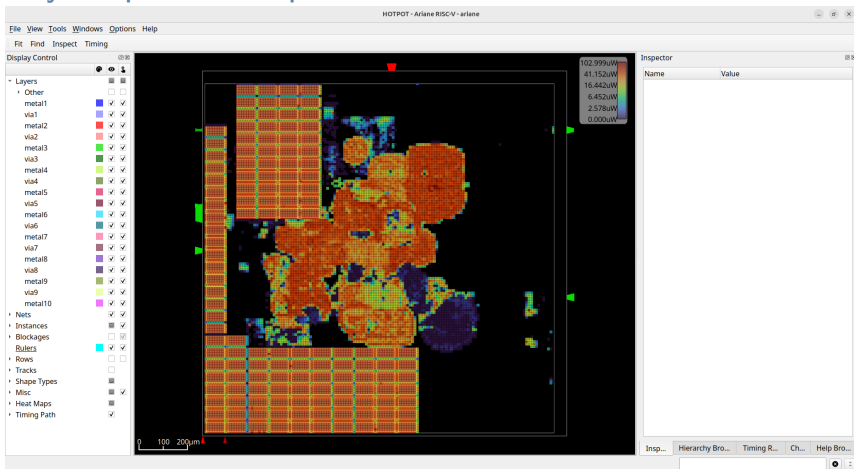
Inspector

Name	Value
Type	Net
Name	alu_valid_id_ex
Block	ariane
Signal type	SIGNAL
Source type	NONE
Wire type	ROUTED
Special	False
Dont Touch	False
I/T terms	2 Items
1	ex_stage_17_05431_1A
2	issue_stage_16_issue_read_operands_2341_1Q
B/T terms	0 Items
Buffer tree	alu_valid_id_ex
BBox	(804.51, 639.985), (805.79, 641.05)
BBox Wldt...	(1.28, 1.065)

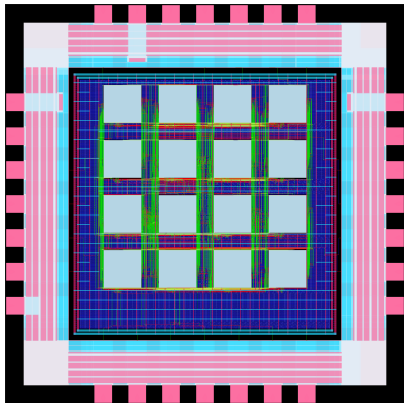
BTerm: a top-level port



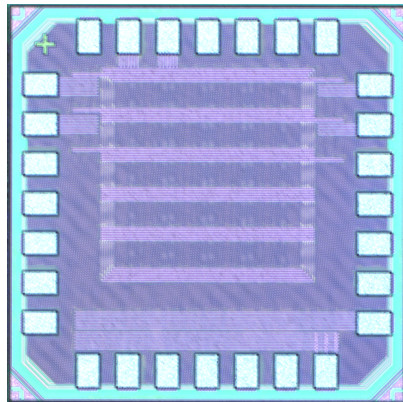
Power density: explore the spatial variation



Frida: floorplan and fabricated silicon



IHP130 layout



Fabricated TSMC65 die

Similar architecture; different processes and geometries.

The command reference

Database
GUI
Web Viewer
Synthesis
Partition Manager
Restructure
Floorplan Initialization
Pin Placement
Design for Test
Chip-level Connections
Macro Placement
Tapcell Insertion
PDN Generation
Global Placement
Static Timing Analyzer
(external)
Parasitics Estimation
Gate Resizing
Clock gating
Detailed Placement



Options

Switch Name	Description
<code>-core_area</code>	Core area coordinates in microns. Either a rectangle as <code>{llx lly urx ury}</code> , or a rectilinear polygon as a list of vertex coordinates <code>{x1 y1 x2 y2 ...}</code> . See Rectilinear Floorplans .
<code>-core_space</code>	Space around the core in microns. Allowed values are either one value for all margins or a set of four values, one for each margin. The order of the four values are: <code>{bottom top left right}</code> .
<code>-die_area</code>	Die area coordinates in microns. Either a rectangle as <code>{llx lly urx ury}</code> , or a rectilinear polygon as a list of vertex coordinates <code>{x1 y1 x2 y2 ...}</code> . See Rectilinear Floorplans .
<code>-site</code>	Site name.
<code>-utilization</code>	Percentage utilization. Allowed values are <code>double</code> in the range <code>{0-100}</code> .
<code>[-additional_sites]</code>	Tcl list of sites to make rows for (e.g. <code>{SITEXX SITEYY}</code>)

Start the workshop script

```
export OPENROAD_ROOT=/path/to/OpenROAD  
openroad -gui flow.tcl
```

Run from the public repository root.

Read the physical library

```
read_lef $tech/Nangate45_tech.lef
read_lef $tech/Nangate45_stdcell.lef
Technology    Cell masters
```

Read the timing library

```
read_liberty $tech/Nangate45_typ.lib  
1.1 V      25 °C
```

Read and link the netlist

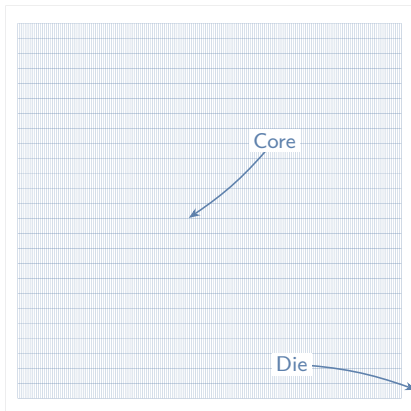
```
read_verilog $netlist
```

```
link_design gcd
```

Top module: gcd

Floorplan comes next

Initialize the floorplan



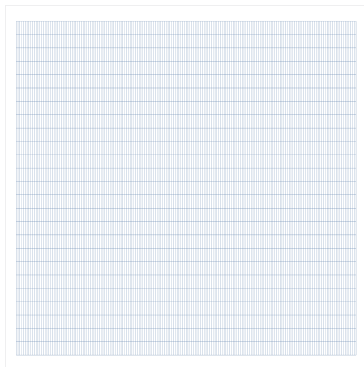
```
initialize_floorplan  
  -site <site>  
  -utilization <percent>  
  -core_space <micrometres>  
  [-aspect_ratio <ratio>]
```

Default: ratio 1.0

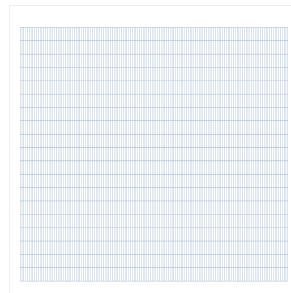
Shown: 40%, margin 1 μm

[argument] optional
<value> placeholder

Utilization sets the available area



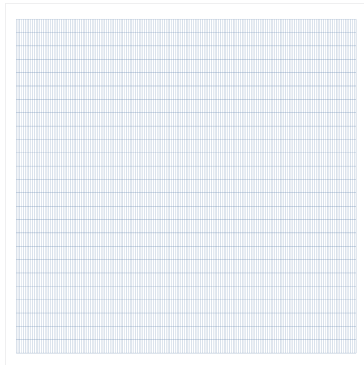
-utilization 40



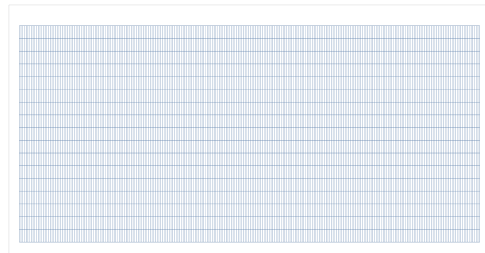
-utilization 65

$$A_{\text{core}} \approx A_{\text{cells}} / (u/100)$$

Choose the shape



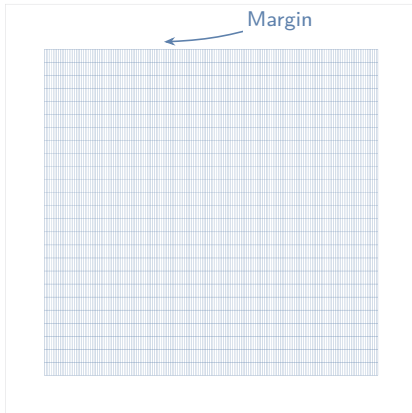
`-aspect_ratio 1`



`-aspect_ratio 0.5`

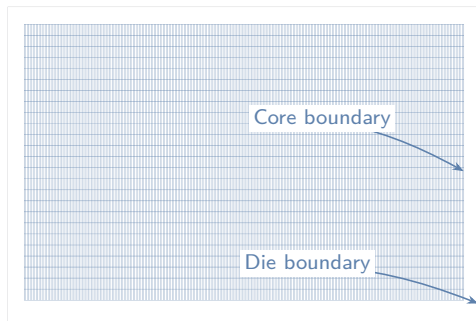
$$r = H/W \quad W = \sqrt{A/r} \quad H = \sqrt{Ar}$$

Reserve the core margin



```
initialize_floorplan  
-site $site  
-utilization 40  
-core_space 4
```

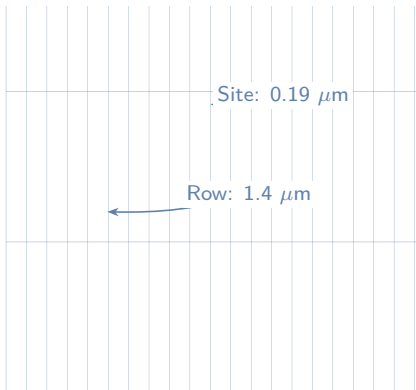
Use a prescribed footprint



```
initialize_floorplan
-site $site
-die_area {0 0 60 40}
-core_area {2 2 58 38}
```

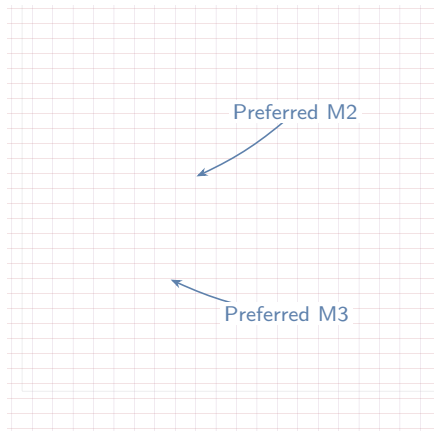
Coordinates in μm

Rows and sites



```
initialize_floorplan  
-site $site  
-utilization 40  
-core_space 1
```

Create routing tracks



`make_tracks`

Pitch and offset from technology LEF

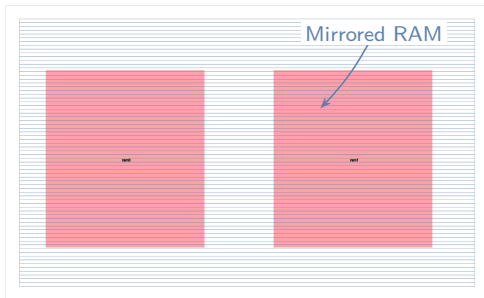
A separate macro example



```
openroad -gui <macros.tcl>
```

Fresh session: two RAM abstracts + GCD

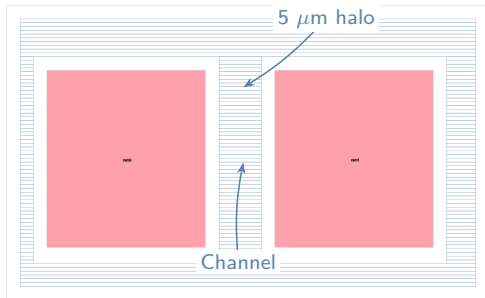
Place and orient a macro



```
place_macro  
  -macro_name ram1  
  -location {100 20}  
  [-orientation MY]
```

Default orientation: R0

Leave access around the macros



```
cut_rows  
[-halo_width_x 5]  
[-halo_width_y 5]
```

Automatic macro placement



`rtl_macro_placer`

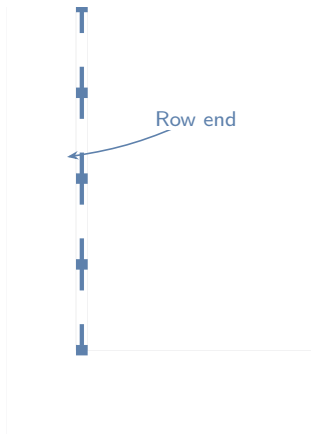
`macros_auto.tcl`: 5 μ m halos

Hierarchical clustering

Simulated annealing

Wirelength + outline + packing

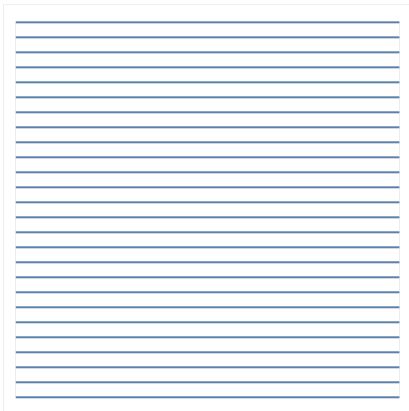
Insert taps and endcaps



```
tapcell  
  [-distance 120]  
  [-tapcell_master TAPCELL_X1]  
  [-endcap_master TAPCELL_X1]
```

GCD branch

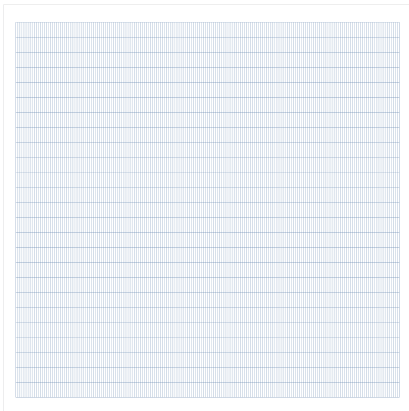
Connect the supply pins



```
add_global_connection  
-net VDD  
-pin_pattern {^VDD$}  
[-power]
```

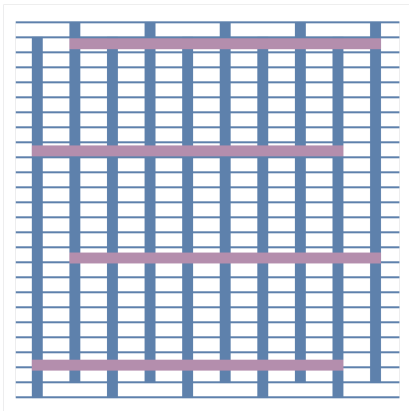
Repeat for VSS with -ground

Define the power domain



```
set_voltage_domain  
-name CORE  
-power VDD  
-ground VSS
```

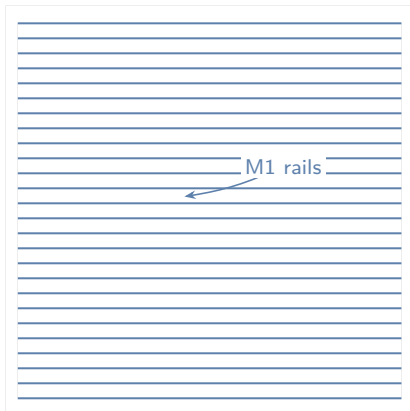
Define the grid



```
define_pdn_grid  
  [-name core_grid]  
  [-voltage_domains CORE]
```

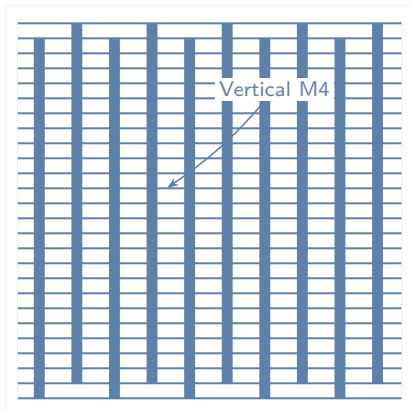
Grid policy; geometry at pdngen

Follow the row power rails



```
add_pdn_stripe  
[-grid core_grid]  
-layer metal1  
[-followpins]
```

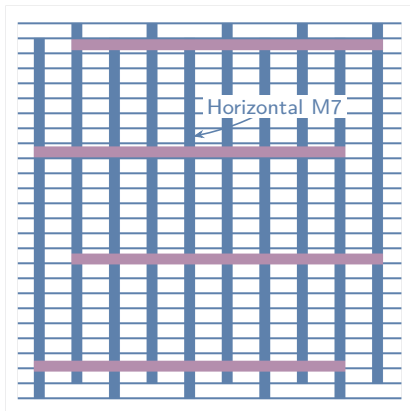
Choose strap width and pitch



```
add_pdn_stripe  
  [-grid core_grid]  
  -layer metal4  
  [-width 1]  
  [-pitch 7]  
  [-offset 2]
```

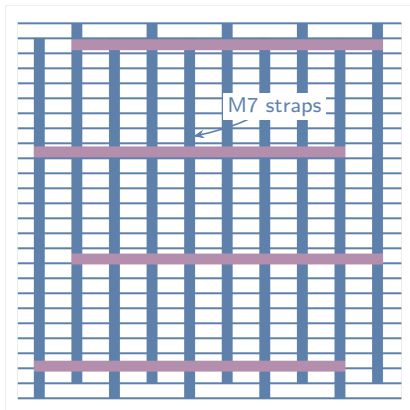
Pitch repeats a VDD/VSS pair

Add a crossing strap layer



```
add_pdn_stripe
-layer metal7
[-width 1]
[-pitch 20]
[-offset 3]
[-number_of_straps 2]
```

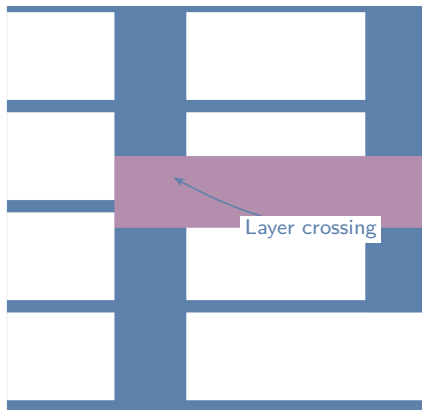
Connect the power layers



```
add_pdn_connect  
  [-grid core_grid]  
  -layers {metal1 metal4}
```

Also connect M4 to M7

Generate the power grid



pdngen

Rule-based straps and via insertion

When do we place the pins?

Pins first

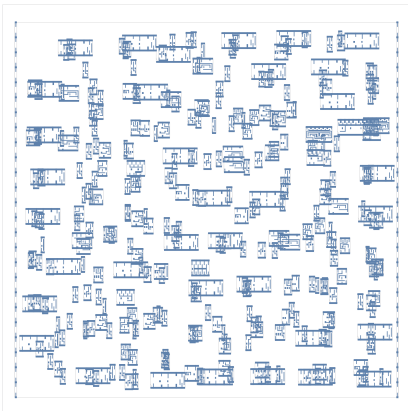
Pin constraints → pin locations → cell placement

Wirelength-driven pin locations

Rough cells → optimize pins → refine cells

Unplaced cell endpoints are modeled at the die center.

Optional: rough cells before pin optimization



```
global_placement  
[-density 0.80]  
[-pad_left 2]  
[-pad_right 2]  
[-skip_io]
```

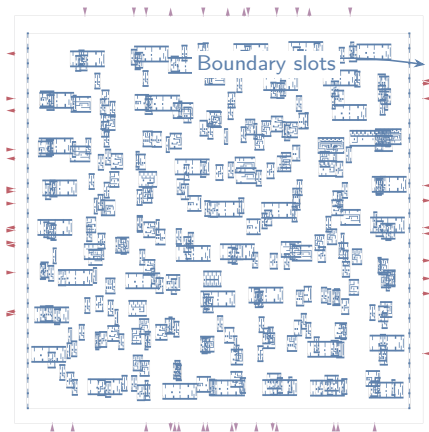
Rough global placement; detailed placement comes later

Smooth wirelength + electrostatic density

Nesterov optimization

Defaults: density 0.7; padding 0

Automatic pin placement



```
place_pins
```

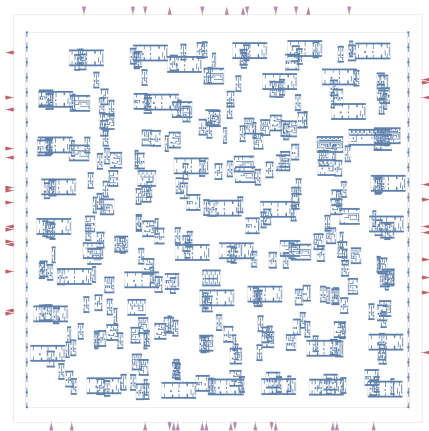
```
-hor_layers metal3
```

```
-ver_layers metal2
```

Fixed cell locations + track slots

Unplaced cells: modeled at die center

The pin-assignment objective



$$c_{ps} = \text{HPWL}(S_p \cup \{s\})$$

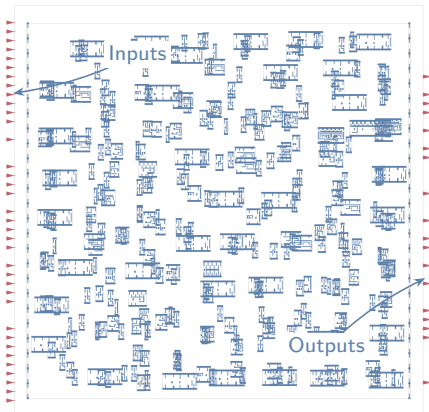
S_p : fixed cell-pin locations

Hungarian matching within slot sections

Optional simulated annealing

Geometric cost; no routed paths

Restrict pins to an edge

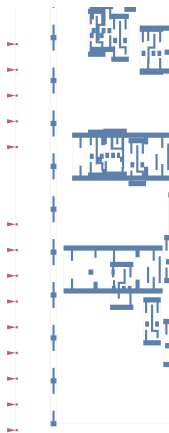


```
set_io_pin_constraint  
  [-direction input]  
  [-region left:*
```

Outputs: -region right:*

Then rerun place_pins

Control the minimum separation



```
place_pins
```

```
-hor_layers metal3
```

```
-ver_layers metal2
```

```
[-min_distance 0.8]
```

0.8 μ m minimum; track snapped

-min_distance_in_tracks: track count

Default separation: two routing tracks

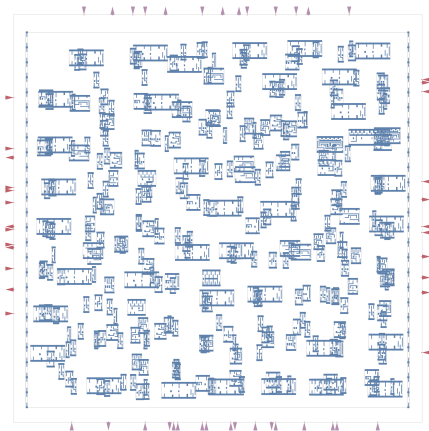
Keep a bus together



```
set_io_pin_constraint
[-pin_names {resp_msg[*]}]
[-region right:*)
[-group]
[-order]
```

Then rerun place_pins

Try the annealing solver

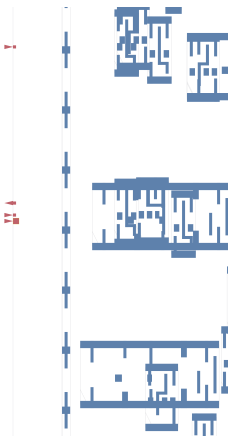


```
place_pins  
  -hor_layers metal3  
  -ver_layers metal2  
  [-annealing]
```

Same HPWL objective

Try edge + bus constraints

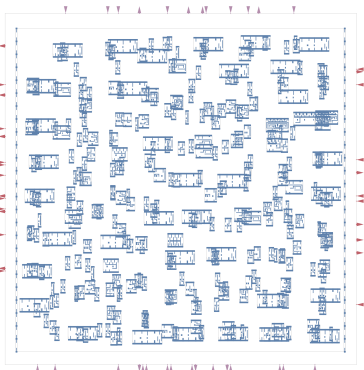
Fix one pin first



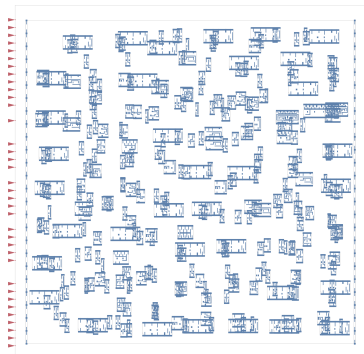
```
place_pin
  -pin_name clk
  -layer metal3
  -location {0 10}
  [-force_to_die_boundary]
```

Then place_pins for the remaining ports

Measure the cost of constraints



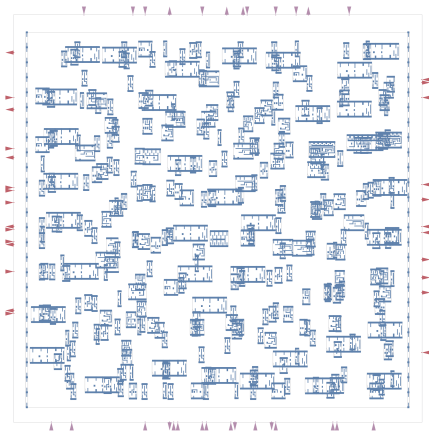
Automatic: 536.82 μm



Constrained: 1126.27 μm

Same fixed cells; constraints: opposite edges + 0.8 μm spacing

Refine cells with the pins included



```
global_placement  
  [-routability_driven]  
  [-density 0.80]  
  [-pad_left 2]  
  [-pad_right 2]
```

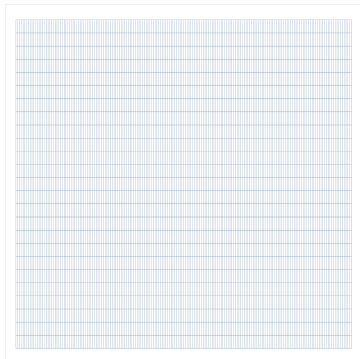
Pins now contribute to the placement objective

RUDY congestion estimate + cell inflation

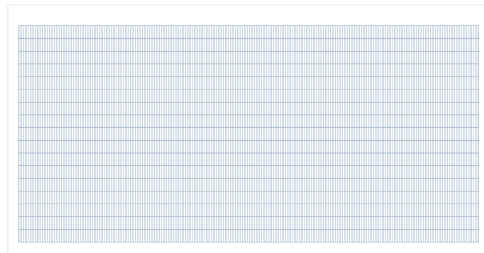
-routability_use_grt: FastRoute feedback

Defaults: density 0.7; padding 0

Exercise: compare two floorplans



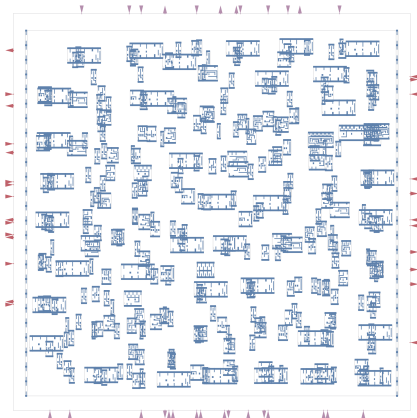
$$u = 40\%, H/W = 1$$



$$u = 40\%, H/W = 0.5$$

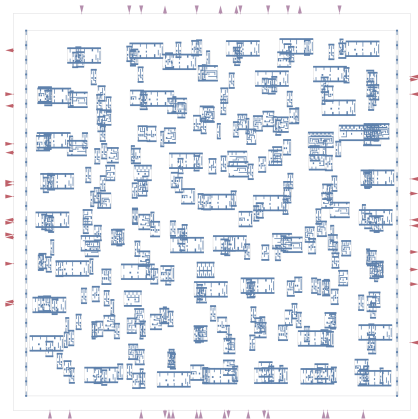
Compare I/O HPWL and pin constraints after placement

Save a half-finished design: ODB



```
write_db  
    partial.odb  
Physical database checkpoint
```

Resume the ODB in a fresh session



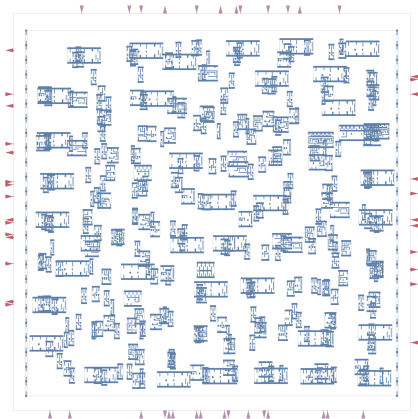
```
read_db
```

```
    partial.odb
```

Technology + cell masters are included

Timing: also read Liberty + SDC

Save a half-finished design: DEF

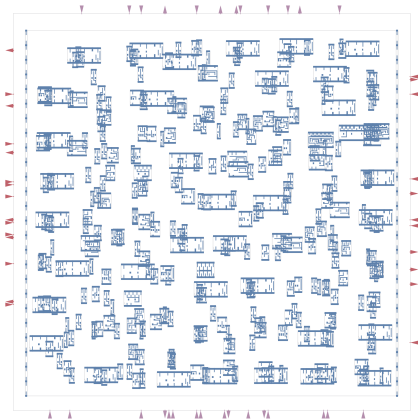


```
write_def
```

```
    partial.def
```

Current rows, pins and placement; routes if present

Read the DEF in a fresh session



```
read_lef $tech/Nangate45_tech.lef  
read_lef $tech/Nangate45_stdcell.lef
```

```
read_def  
    partial.def
```

LEFs supply technology + cell masters
Timing: also read Liberty + SDC

Open the workshop checkpoints

```
openroad -gui examples/floorplanning/read_odb.tcl  
openroad -gui examples/floorplanning/read_def.tcl  
Separate sessions; run from slides/.
```

```
cd examples/ariane  
make run  
make gui
```

Ariane: pinned downloads + standalone OpenROAD.